Introduction -
In order to acquire our requirements we met up with our customer and interviewed them since there was no pre existing system to base our requirements off of. We prepared a few questions beforehand grouped by themes of the gameplay, difficulty and accesibility to ensure that we have a wide coverage of all possible areas before we start.
Multiple different requirements were refined through the interview, some initial ideas of a health bar or patience meter were constrained to being a single hunger bar. In addition we fixed the scope of the project by constraining the time limit to 5 minutes and establishing some ideas about the story and visual design of the game, making sure to not assume anything about the project and instead agreeing with the customer to correctly identify the boundaries. (Nuseibeh and Easterbrook)

In order to display and keep track of our requirements easily we have chosen to display them in tables with an order of priority so we can make sure we include the main requirements per our customer's needs. Simplified IDs are used to easily summarise the requirements whilst the description goes more into more detail about what the requirement entails.

User Requirements

ID	Description	Priority
UR_AVATAR	The player shall control a character avatar representing a University of York undergraduate student	Shall
UR_CAMPUS_MAP	The game shall contain a map of the university campus with clear limits on where the player can go with obvious landmarks and locations to indicate that it is the UoY campus.	Shall
UR_GEESE	The game shall contain an onslaught of geese demanding food, which prevent the player from simply running away	Shall
UR_ITEMS	The player shall start with an inventory of at least five with five more items places around the map to smartly manage the geese, with some items being multi use or one time consumables.	Shall
UR_EVENT_COUNT	The game shall contain at least five hidden events, invisible until triggered, with positive or negative outcomes	Shall
UR_SCORE	The player shall receive a score based on how quickly the situation is resolved, how smartly items are used and how hidden events are handled	Shall
UR_PAUSE	The game shall start paused and shall allow the player to pause at any time (at least on base difficulty)	Shall
UR_NO_VIOLENCE	The game shall not contain killing, invasion or violence, and the player shall not be able to fight the	Shall

	geese	
UR_TIME_LIMIT	The game shall end after a hard cap of 5 minutes	Shall
UR_WIN_LOSE	The player shall win by removing at least the main horde before time runs out or else they shall lose	Shall
UR_HUNGER_BAR	The game shall use hunger bars rather than health bars	Shall
UR_STAMINA_BAR	The game may include a stamina bar	May
UR_NPC	The game should include NPCs that handle the random and hidden events, placed around the map with obvious markers to indicate where they are.	Should
UR_DIFFICULTY	The game should offer multiple difficulty levels, where higher levels change some indicators with the base difficulty being non-punishing.	Should
UR_LEADERBOARD	The game may store player scores and display them on a leaderboard	May
UR_SIMPLE_STORY	The game shall keep the storyline simple and shall not include a full storyline	Shall
UR_ACCESSIBILITY	The game should accommodate for less abled people so that they can have the same experience	Should

Functional Requirements

ID	Description	User Requirements	Priority
FR_MOVEMENT	The system shall allow the player to move the avatar around the map using keyboard controls	UR_AVATAR	Shall
FR_MAP_BOUNDARIES	The system shall prevent the player from moving outside the map boundaries or through obstacles	UR_CAMPUS_MAP	Shall
FR_GEESE_SPAWN	The system shall spawn geese of varying sizes that move towards the player	UR_GEESE	Shall
FR_GEESE_BLOCK	The system shall make geese obstruct the player's movement when they are near the player	UR_GEESE	Shall
FR_GOOSE_HUNGER	The system shall give each goose a hunger bar and once it is filled it will leave.	UR_GEESE UR_WIN_LOSE	Shall

FR_INVENTORY	The system shall have an inventory of at least 5 spaces to include the starter items to accommodate the pickups placed around the map	UR_ITEMS	Shall
FR_USE_ITEM	The system shall allow the player to use an inventory item to interact with the geese with each item having a different effect	UR_ITEMS UR_SCORE	Shall
FR_ITEM_INDICATORS	The system should mark items on the map and threats with clear visual indicators	UR_ACCESSIBILITY	Should
FR_STAMINA	The system should limit sprinting with a stamina bar that drains when sprinting and recovers when resting	UR_STAMINA_BAR	Should
FR_HIDDEN_EVENTS	The game shall contain at least five hidden events, activated upon reaching or interacting with specific locations or NPCs and will provide a beneficial or detrimental result upon activating a secret event, ultimately rewarding the player once finished	UR_HIDDEN_EVENTS UR_EVENT_COUNT	shall
FR_TIMER	The system shall display a countdown of the 5-minute time limit and end the game when it reaches zero	UR_TIME_LIMIT UR_NO_EXTENSION	Shall
FR_WIN_LOSE	The game will end in victory if the main horde of geese are dispersed within the time limit, if not then the user loses.	UR_WIN_LOSE	Shall
FR_DIFFICULTY_SELECT	The game should offer a difficulty selection option at the start.	UR_DIFFICULTY	Should
FR_SCORE_CALC	The system shall calculate a score from the time taken, item use and hidden events handled	UR_SCORE, UR_POINTS	Shall
FR_COUNTERS	The system shall display counters of items used and events triggered	UR_ITEMS UR_HIDDEN_EVENTS	Shall
FR_COLOUR_MODES	The system should accommodate for colour blindness, allowing the user to pick between colour schemes	UR_ACCESSIBILITY	Should
FR_AUDIO_CUES	The system should play audio cues to notify the user in case they miss the visual cue	UR_ACCESSIBILITY	Should